

PRESENTED FOR EVALUATION

AI-Powered Multilingual News Portal

with a DevOps CI/CD Pipeline

Git · GitHub · Jenkins · Docker · Node · React · MongoDB Atlas · Gemini AI

TEAM

TBD

PROGRAM

TBD

INSTITUTION

TBD

DATE

TBD

WHY THIS PROJECT

Regular news sites fall short in three ways.

01

One language only

Most Indian regional readers are excluded from English-first news portals.

02

No read-aloud

Zero accessibility for low-literacy or visually impaired readers.

03

Raw agency copy

Unedited wire copy — verbose, inconsistent tone, no editorial polish.

Our answer: AI rewriting · contextual AI translation · browser-native text-to-speech.

GOAL

Ship a real app through a real DevOps pipeline.

THE APP

A multilingual news portal with AI rewriting, contextual translation, and text-to-speech.

Realistic, non-trivial workload for the pipeline to build & deploy.

THE DEVOPS FOCUS (EVALUATION)

Git · GitHub · Jenkins · Docker · CI/CD
— a hands-off pipeline from **git push** to a running container.

Zero manual steps between commit and deploy.

WHAT WE USED

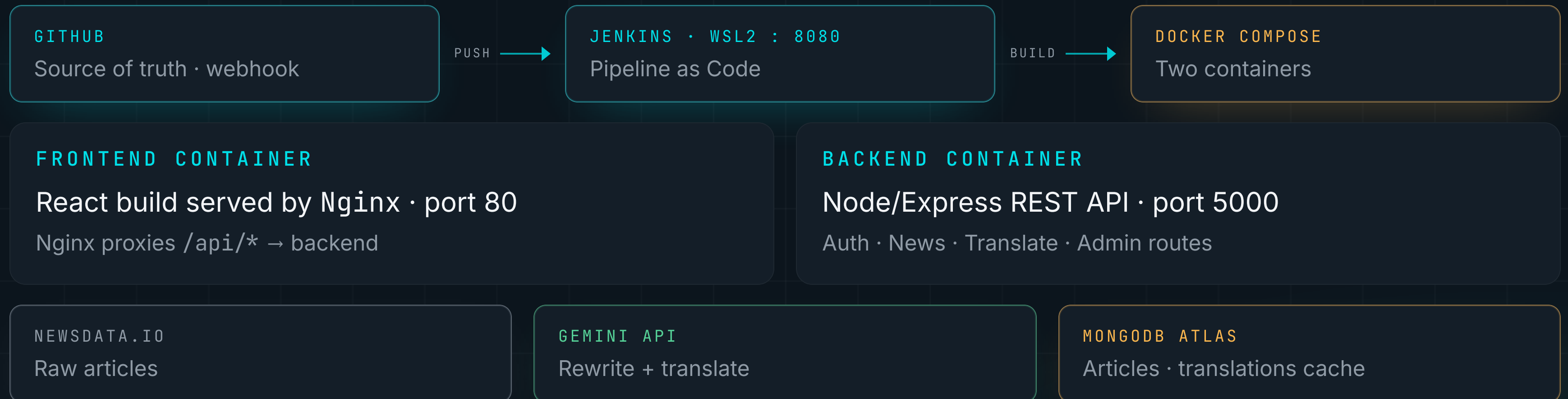
Tech Stack Overview

LAYER	CHOICE	WHY
Frontend	React + Vite	Fast dev/build tooling, modern DX
Backend	Node.js + Express	Lightweight REST API
Database	MongoDB Atlas (M0)	Flexible schema, cloud-hosted, free
AI	Google Gemini 1.5 Flash	Rewrite + contextual translation
News Source	NewsData.io API	Reliable India-focused feed
Auth	JWT (admin only)	Stateless, simple, standard
Containers	Docker + Compose	Reproducible build & deploy
Reverse Proxy	Nginx (in frontend)	Routes /api → backend
CI/CD	Jenkins (Pipeline)	Jenkinsfile as code
VCS	Git + GitHub webhook	Auto-triggers Jenkins
Dev Env	WSL 2 (Ubuntu)	Linux tooling on Windows

HOW IT FITS TOGETHER

System Architecture

Push to GitHub triggers Jenkins, which builds and deploys the app via Docker Compose.



One document holds source, AI output, and every cached translation.

CORE FIELDS

<code>title / content / summary</code>	AI-rewritten, shown by default
<code>originalTitle / originalContent</code>	Untouched source, kept for reference
<code>imageUrl · videoUrl</code>	Optional embedded video
<code>source { name, url }</code>	Attribution
<code>category</code>	9 categories incl. India, World, Tech
<code>status</code>	draft · published (admin gate)

THE CLEVER BIT

```
translations: Map<langCode,  
  { title, content, summary }  
>
```

Each language is translated by Gemini **once**, then cached forever. Repeat views cost nothing.

PUBLIC SIDE

Key Features — User Side

- Newspaper-style homepage — featured story + card grid
- Category browsing across 9 sections
- Full article page with image / embedded video
- Language selector — **10 languages** incl. Hindi, Marathi, Tamil, Bengali, French
- **Listen** button — browser-native text-to-speech, zero cost
- Responsive: newspaper grid on desktop, stack on mobile

SCREENSHOT

Homepage grid view

replace with real image

SCREENSHOT

Article page with translated language selected

replace with real image

EDITORIAL SIDE

Key Features — Admin Side

Editorial approval workflow — nothing publishes without a human.

- Secure JWT login — credentials from env vars, never hardcoded
- Dashboard stats — total / published / draft / AI-rewritten counts
- Manual **Fetch News** by category → NewsData → Gemini rewrite → save as draft
- Draft review queue — Preview · Publish · Delete
- Published article management
- Admin approves before public — real newsroom pattern, not autoposting

SCREENSHOT

Admin dashboard with stats

replace with real image

GEMINI 1.5 FLASH

How AI powers the portal

Two focused functions — one rewriter, one translator — plus a summary helper.

`rewriteArticle()`

Neutral journalistic tone

Rewrites raw wire copy into a 200–300 word neutral piece. Facts preserved, opinions stripped.

`translateArticle()`

Contextual translation

Not word-for-word. Keeps proper nouns intact. Returns strict JSON for clean caching.

`generateSummary()`

2-sentence preview

Short summaries shown on the homepage newspaper grid cards.

TRANSLATION FLOW • CACHING KEEPS IT FAST & CHEAP

POST /api/translate → Cache hit? `yes` → `return` • `no` → `Gemini` → `cache` → `return`

TRADE-OFFS MADE INTENTIONALLY

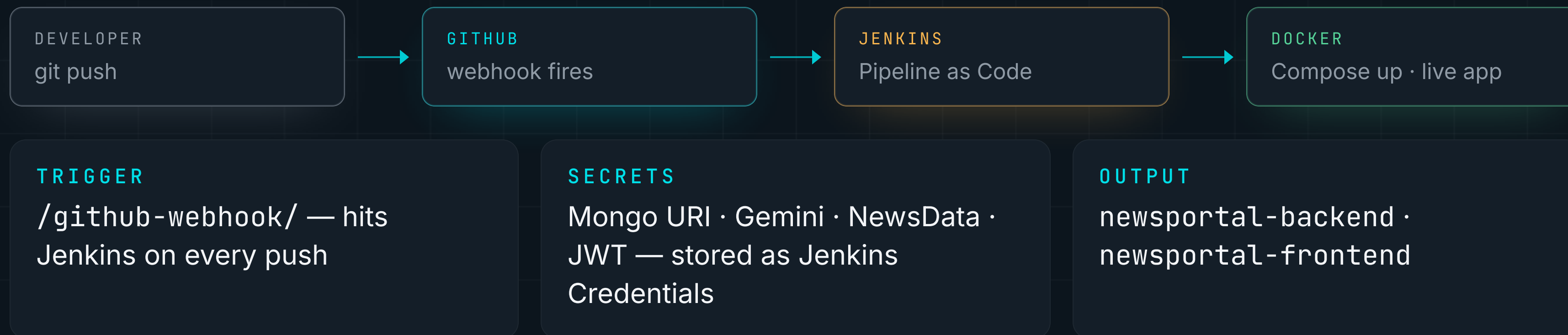
Key Design Decisions & Rationale

DECISION	REASONING
Admin-triggered fetch, not cron	No 24/7 uptime needed for demo · mirrors real editorial approval
Translation caching in MongoDB	No repeat Gemini calls for same (article, language) pair
MongoDB Atlas over local DB	Data persists when laptop is off · free tier forever
Browser speechSynthesis over paid TTS	Zero cost, instant, multi-language built in
NewsData.io, not GNews	GNews free tier returned 403 on server-side calls
Vite, not Create React App	CRA repeatedly broke on install · Vite is faster & maintained
Jenkins WAR, not apt install	Ubuntu 26.04 GPG key issues with Jenkins apt repo
No Kubernetes / Prometheus / Grafana	Not yet in training · unnecessary at this scale

ZERO MANUAL STEPS

Git → GitHub → Jenkins → Docker

Every commit auto-triggers the pipeline via GitHub webhook.



Jenkinsfile — 9 stages, top to bottom

01	Checkout	pull latest from GitHub
02	Install Backend Deps	npm install in backend/
03	Install Frontend Deps	npm install in frontend/
04	Test	run backend tests
05	Build Frontend	npm run build · production React
06	Build Docker Images	docker compose build
07	Stop Old Containers	docker compose down
08	Deploy	compose up -d · secrets injected

SCREENSHOT

Jenkins pipeline stage view · all green

replace with real image

WHAT BROKE – AND HOW WE FIXED IT

Challenges Faced & Solutions

PROBLEM	CAUSE	FIX
MongoDB crypto not defined	Node 18 incompatibility	Upgraded to Node 20
Jenkins apt install failed	Ubuntu 26.04 GPG / NO_PUBKEY	Installed via .war + java -jar
Jenkins needs Java 21	Only Java 17 present	OpenJDK 21 · update-alternatives
GNews API 403 Forbidden	Free tier blocks server calls	Switched to NewsData.io
react-scripts not found	CRA install broken in WSL2	Rebuilt frontend with Vite
Corrupted JSX on paste	Terminal paste buffer in nano	Used heredoc (cat > file << EOF)
Admin fetch: undefined articles	Response shape changed with new API	Debugged with curl · fix applied · re-test pending

Project Status

✓ CONFIRMED WORKING

Backend API — auth · news ·
translate · admin

MongoDB Atlas connected &
tested

React/Vite frontend — all pages
functional

Admin login & JWT flow

Text-to-speech "Listen" feature

Jenkins running locally,
credentials configured

🕒 CODED • NEEDS RE-VERIFY

AI rewrite + translation via Gemini
— key needs final verification

NewsData.io fetch integration —
fix applied, end-to-end run
pending

Stated plainly — not claiming "fully
working" until confirmed live.

🔄 IN PROGRESS

Git init + first GitHub push

Dockerfile build testing

docker-compose end-to-end run

Jenkins Pipeline job + webhook
wiring

Full push-to-auto-deploy demo

HOW WE'LL PRESENT IT LIVE

Demo Plan

A single push visibly triggers the entire pipeline end-to-end.

01 Show GitHub repo — code, Jenkinsfile, Dockerfiles, docker-compose

03 Show Jenkins dashboard — pipeline auto-triggers, all stages run live

05 Open the live site — confirm the change appears

07 Open article · switch to Hindi · click Listen · TTS reads aloud

02 Make a small visible code change, commit & push

04 Run docker ps — both containers up

06 Log into /admin · click Fetch News · show AI rewriting · publish drafts

08 Hit /health endpoint — health-check demonstration

Conclusion & Learnings

KEY LEARNINGS

- Pipeline as Code — Jenkinsfile in source control
- Docker & Compose for reproducible multi-service deploys
- Webhooks & true CI/CD — no manual steps between push and prod
- Secrets discipline — env-injected credentials, never in Git
- Practical AI integration behind a clean caching layer

FUTURE SCOPE

- **Kubernetes** — orchestration, scaling, self-healing
- **Prometheus + Grafana** — metrics & dashboards
- **Cloud VM** — 24/7 hosting beyond the local demo
- Automated news fetch via scheduled jobs

Thank you — questions welcome.